18 $\quad f(x) = 2x + 5$

$$f(x)f(x) - ff(x) = ax^2 + bx + c$$

Find the value of a, the value of b and the value of c.

$(2x + 5)(2x + 5) - 2(2x + 5) + 5$

$2x^2 + 10x + 10x + 25 - 4x - 10 + 5$

$2x^2 + 20x + 25 - 4x - 10 + 5$

$2x^2 + 16x + 20$

$a = $ .... 2 ....

$b = $ .... 16 ....

$c = $ .... 20 .... [4]

19 $\quad$ Solve.

$$\left(\frac{1}{3}\right)^x = 9^{x+4}$$

$\frac{1^x}{3^x} = (3^{-2})^{(x+4)}$

$\frac{1^x}{3^x} = 3^{-2x-8}$

$\frac{1^x}{3^x} = \frac{1}{3^{2x+8}}$

$x = 2x + 8$

$-8 = 2x - x$

$-8 = x$

$x = $ .... -8 .... [3]